

# FABIO CAMPOS VILLAVERDE

Mechanical Engineer · Design & Manufacturing · SolidWorks & Simulation

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## PROFESSIONAL PROFILE

Mechanical Engineer from the University of Vigo (specialization in Design & Manufacturing), graduating in September 2026. I take projects from parametric CAD all the way to a functional prototype, with hands-on experience in 3D modeling (SolidWorks and Fusion 360), additive manufacturing and finite element analysis (FEA). I'm passionate about mechanical product design, mechanisms and real-world fabrication: I enjoy solving engineering problems end to end, combining technical judgment with a strong maker culture and self-driven learning.

## EDUCATION

### BSc in Mechanical Engineering — University of Vigo

*Expected graduation: Sept. 2026*

*Specialization in Design & Manufacturing · Only the Bachelor's Thesis (TFG) remaining*

- **Bachelor's Thesis:** design of the optomechanical architecture and structural support of the light-detection modules (SiPM sensors) of the ND-GAr near detector for the international DUNE experiment. Complete CAD model in SolidWorks with a modular sub-assembly architecture (400×400 mm tileable envelope). [View project ↗](#)
- **Cryogenic challenge ( $\Delta T = -47\text{ °C}$ ):** kinematic isolation design with slotted holes, PEEK spacers and Belleville washers to absorb ~1 mm of differential contraction without damaging the sensors; analytical sizing of the bolted joint (Rötscher criterion) and groundwork for thermomechanical validation by FEA.
- **Notable grades:** Outstanding (9.0/10) in Data Analysis and in Simulation & Validation Systems. Strong foundation in Technical Drawing and Machine Design.

### University Certificate in Computer-Aided Design — Advanced SolidWorks (University of Vigo)

## TECHNICAL SKILLS

- **3D Modeling & CAD:** SolidWorks and Fusion 360 (advanced — parametric design, sheet metal, large assemblies and drawings), AutoCAD (intermediate), CATIA (basic). Quick to pick up new software (e.g. Inventor).
- **Manufacturing:** additive manufacturing / 3D printing (own 3D printer for years; functional part design), basic CNC machining, reverse engineering and structural assembly.
- **Simulation (CAE):** finite element analysis (FEA) and structural calculation with ANSYS and SolidWorks Simulation.
- **Programming & electronics:** Python (data analysis), C++ (Arduino / ESP32), IoT and AI/LLM integration.
- **Languages:** Spanish and Galician (native) · English (B1 certified; working proficiency B2–C1).

## ENGINEERING PROJECTS

### Modular Aggregate Conveyor Belt — Parametric Design (SolidWorks)

[view project ↗](#)

- Modeled in SolidWorks (part, sheet metal and top-down assemblies), parametrized with global equations and design tables: a replicable “base module” that automatically adjusts length and the number of rollers and hardware without redesigning parts.
- Integration of commercial components (Toolbox) and guards per machine-safety standards.
- Manufacturing documentation and photorealistic (PBR) rendering in SolidWorks Visualize. Completed as the project for my advanced-CAD university certificate.

### Direct Drive Sim Rig 15 Nm — Mechatronics & Reverse Engineering

[view project ↗](#)

- 15 Nm Force Feedback wheel built from a repurposed motor (reverse engineering) and the open-source FFBeast ecosystem; most parts were designed in CAD and produced by 3D printing (housing, cooling and functional supports).
- Pedals with hydraulic feel and a drift handbrake using 100 kg load cells and elastomers; sequential shifter with a custom cam mechanism.

- Telemetry integration (SimHub) with LED matrices and haptic bass shakers. A complete system designed, built and fine-tuned end to end.

#### **IoT Hangboard — Electronics & Data Acquisition**

[view project ↗](#)

- Precision training board with an industrial-grade load cell and an ESP32, emulating the BLE protocol of commercial sensors (compatible with third-party apps).
- Mechanical design with interchangeable 3D-printed inserts for different edge depths.
- Custom mobile app with real-time force visualization (50 Hz); I iterated several sensor versions until achieving reliable measurements.

### **EXPERIENCE**

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#### **Construction & Maintenance Worker — Campos Núñez S.L.**

*Summers 2018–2023*

- Hands-on experience in structural assembly and on-site technical execution in demanding environments.

#### **Climbing Coach (volunteer) — Grupo de Montaña Alud**

*Present*

- Training competitive athletes at regional level; planning, discipline and teamwork.

### **INTERESTS**

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- Climbing & mountaineering: a sport I practice and coach at competitive level; it gives me discipline and a problem-solving mindset.
- Maker culture & 3D printing: I design and build my own projects as a way to learn, with a growing interest in AI applied to engineering.